



Modelling Financial Markets

Seminar 7: Value-at-Risk and Expected Shortfall — Hands-on



Daniel Traian PELE

Drd. Antoaneta Amza, antoaneta.amza@csie.ase.ro

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CF: Cornish–Fisher (expansion)
- ▣ CI: Confidence Interval
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ ES: Expected Shortfall
- ▣ EVT: Extreme Value Theory
- ▣ FRTB: Fundamental Review of the Trading Book
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GPD: Generalised Pareto Distribution
- ▣ OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))
- ▣ SE: Standard Error
- ▣ XFG: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

Outline

Introduction

Part A: Pen and Paper

Part B: Estimation and Inference

Part C: Open Question

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- Compute VaR and ES for the Normal and Student- t distributions step by step
 - Show why VaR can fail subadditivity and ES cannot
 - Estimate historical, filtered and extreme-value VaR/ES with confidence intervals
 - Split portfolio risk into Euler contributions
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- Background: Lecture 7; textbooks McNeil, Frey & Embrechts (2015) and Franke, Härdle & Hafner (2019)

Structure

- **Part A:** six pen-and-paper exercises (A1–A6)
- **Part B:** six computational exercises with inference (B1–B6)
- **Part C:** one open question, a seed for the team project (C1), and a critique of an AI answer (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- Sign convention: loss $L = -r$ in % of the position; VaR and ES are positive numbers
- Report every tail estimate with its uncertainty: how many observations lie beyond it?

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (Normal VaR and ES, step by step), A2 (Student- t VaR and ES, step by step), A3 (a discrete loss), A5 (component VaR of two assets)
- Part B: B1 (historical VaR and ES on BET with a bootstrap interval), B2 (GPD fit and extreme-value VaR on BET)

Proposed: you solve them

- Part A: A4 (failure of subadditivity; model: A3), A6 (Cornish–Fisher; model: A1)
- Part B: B3 (filtered vs historical on Bitcoin; model: B1), B4 (Cornish–Fisher vs empirical; model: A1, B1), B5 (component VaR of a portfolio; model: A5), B6 (bootstrap interval for ES; model: B1)
- Part C: C1, open question and project seed; C2, critique of an AI answer (model: B1)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter7_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `arch`, `matplotlib`
- Series: BET (2000–2026), S&P 500, Bitcoin (2014–2026), EUR/RON (BNR reference rate), gold; SPY, TLT, GLD; eight BVB blue chips; daily closes up to 18 September 2026
- Equity indices: weekdays only, days with an unchanged close removed; crypto keeps all 7 days of the week
- Portfolios: prices joined on common trading days first, then simple returns

Formulas Used Today

- Level α = tail probability (VaR 1%, ES 2.5%); return X , loss $L = -X$: $\text{VaR}_\alpha(X) = -\inf\{x : P(X \leq x) > \alpha\} = -q_\alpha(X)$,
 $\text{ES}_\alpha(X) = -\frac{1}{\alpha} \int_0^\alpha q_u(X) du$
- Discrete case [Acerbi & Tasche \(2002\)](#): $\text{ES}_\alpha = \frac{1}{\alpha} (\mathbb{E}[L1\{L > \text{VaR}_\alpha\}] + \text{VaR}_\alpha(\alpha - P(L > \text{VaR}_\alpha)))$
- Normal distribution, $z_\alpha = \Phi^{-1}(\alpha) < 0$: $\text{VaR}_\alpha = -(\mu + \sigma z_\alpha)$, $\text{ES}_\alpha = -\mu + \sigma \varphi(z_\alpha)/\alpha$
- Student- t : $\text{VaR}_\alpha = -(m + s t_\alpha)$, $\text{ES}_\alpha = -m + s \frac{s\nu(t_\alpha)}{\alpha} \frac{\nu+t_\alpha^2}{\nu-1}$; variance σ^2 when $s = \sigma\sqrt{(\nu-2)/\nu}$
- GPD tail of the losses: $\text{VaR}_\alpha = u + \frac{\beta}{\xi} \left[\left(\frac{n}{N_u} \alpha \right)^{-\xi} - 1 \right]$, $\text{ES}_\alpha = \frac{\text{VaR}_\alpha + \beta - \xi u}{1-\xi}$
- Euler: $C_i = w_i \partial \text{VaR} / \partial w_i = \mathbf{z}_{1-\alpha} w_i (\Sigma w)_i / \sigma_p$, $\sum_i C_i = \text{VaR}$ [Tasche \(2008\)](#)

A1: Normal VaR and ES, Step by Step [Solved]

Task

- Daily returns $X \sim N(0.04\%, 1.2\%^2)$; position 1 million EUR
- (a) VaR 1% and ES 1%, in % and in EUR
- (b) VaR 2.5% and ES 2.5%
- (c) 10-day VaR 1% by the square-root-of-time rule (zero mean)

Solution

- $z_{1\%} = -2.326$, $\varphi(z_{1\%}) = 0.0267$, $\varphi/0.01 = 2.665$
- $\text{VaR}_{1\%} = -(0.04 + 1.2 \cdot (-2.326)) = 2.75\%$ (27 516 EUR)
- $\text{ES}_{1\%} = -0.04 + 1.2 \cdot 2.665 = 3.16\%$ (31 583 EUR)
- $z_{2.5\%} = -1.960$: $\text{VaR}_{2.5\%} = 2.31\%$,
 $\text{ES}_{2.5\%} = -0.04 + 1.2 \cdot 2.338 = 2.77\%$
- $\text{VaR}_{1\%}^{(10)} = -\sqrt{10} \cdot 1.2 \cdot (-2.326) = 8.83\%$ (88 279 EUR)

A2: Student- t VaR and ES, Step by Step [Solved]

Task

- Same mean and standard deviation as A1, but X follows a Student- t distribution with $\nu = 4$
- (a) Scale s that gives a standard deviation of 1.2%
- (b) VaR and ES at $\alpha = 1\%$ and $\alpha = 2.5\%$
- (c) Compare with A1: where do the two models differ most?

Solution

- $s = 1.2\sqrt{2/4} = 0.849$
- $t_{1\%} = -3.747$, $g_4(t_{1\%}) = 0.0087$; factor $\frac{0.0087}{0.01} \cdot \frac{4+14.04}{3} = 5.221$
- $\text{VaR}_{1\%} = -(0.04 + 0.849 \cdot (-3.747)) = 3.14\%$,
 $\text{ES}_{1\%} = -0.04 + 0.849 \cdot 5.221 = 4.39\%$
- $t_{2.5\%} = -2.776$: $\text{VaR}_{2.5\%} = 2.32\%$, $\text{ES}_{2.5\%} = 3.35\%$
- VaR 2.5% almost coincides in the two models (2.32 vs 2.31); for VaR 1% and for ES the t model is much higher: heavy tails show up only deep in the tail

A3: VaR and ES of a Discrete Loss [Solved]

Task

- A bond loses 60 (loss given default) with probability $p = 3\%$ and 0 otherwise: $X = -60$ or $X = 0$
- (a) VaR 5% and VaR 2.5% from the definition
 $-\inf\{x : P(X \leq x) > \alpha\}$
- (b) ES at both levels with the discrete formula
- (c) Why is $ES_{5\%}$ not simply $\mathbb{E}[L \mid L \geq \text{VaR}_{5\%}]$?

Solution

- $P(X \leq -60) = 3\%$, not above 5% $\Rightarrow q_{5\%} = 0$,
 $\text{VaR}_{5\%} = 0$; $3\% > 2.5\% \Rightarrow q_{2.5\%} = -60$, $\text{VaR}_{2.5\%} = 60$
- $ES_{5\%} = \frac{60 \cdot 0.03 + 0 \cdot (0.05 - 0.03)}{0.05} = 36$
- $ES_{2.5\%} = \frac{0 + 60 \cdot (0.025 - 0)}{0.025} = 60$
- $\mathbb{E}[L \mid L \geq 0] = 1.8$ averages over **all** days, not over the worst 5%: with atoms in the distribution the correction term is needed

A4: When VaR Punishes Diversification [Proposed]

Solution discussed in class

Task

- Two independent bonds, each loses 100 with probability $p = 0.7\%$ Artzner et al. (1999)
- (a) VaR 1% and ES 1% of each bond and of the portfolio of both
- (b) Show that VaR is not subadditive and ES is
- (c) Repeat at $\alpha = 1.5\%$: does VaR still fail?
- Model: A3

A5: Component VaR of Two Assets [Solved]

Task

- Equity: 600 000 EUR, $\sigma_1 = 1.2\%$; bonds: 400 000 EUR, $\sigma_2 = 0.9\%$; correlation $\rho = -0.2$; Normal distribution, zero mean
- (a) Portfolio standard deviation and $\text{VaR}_{1\%}$ in EUR
- (b) Marginal and component VaR; check $C_1 + C_2 = \text{VaR}$ Tasche (2008)
- (c) Diversification benefit

Solution

- Stand-alone s.d.: $600\,000 \cdot 1.2\% = 7\,200$,
 $400\,000 \cdot 0.9\% = 3\,600$ EUR;
 $\sigma_p^2 = 7\,200^2 + 3\,600^2 + 2(-0.2)(7\,200)(3\,600) = 54\,432\,000$,
 $\sigma_p = 7\,378$ EUR
- $\text{VaR}_{1\%} = 2.326 \cdot 7\,378 = 17\,163$ EUR
- $(\Sigma w) = (77.76, 19.44)$ EUR; marginal VaR
 $= (0.0245, 0.0061)$
- $C_1 = 14\,711$ (85.7%), $C_2 = 2\,452$ (14.3%); sum = 17 163
- Stand-alone VaRs $16\,750 + 8\,375 = 25\,125$ EUR; bonds hold 40% of the money but only 14.3% of the risk

A6: Cornish–Fisher, Step by Step [Proposed]

Solution discussed in class

Task

- ▣ Returns with $\mu = 0$, $\sigma = 1.2\%$, skewness $S = -0.5$, excess kurtosis $K = 3$ Cornish & Fisher (1938)
- ▣ (a) The three correction terms of $\tilde{z}_{1\%}$ separately, with $z_{1\%} < 0$
- ▣ (b) Cornish–Fisher VaR $1\% = -(\mu + \sigma \tilde{z}_{1\%})$ vs Normal
- ▣ (c) Repeat with $K = 10$ (close to the S&P 500): is the result credible?
- ▣ Model: A1

B1: Historical VaR and ES on BET [Solved]

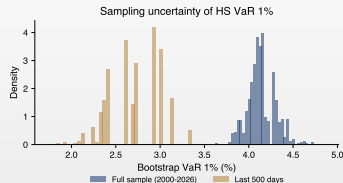
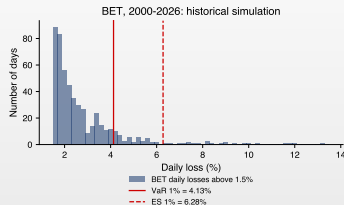
Task

- BET daily losses: full sample 2000–2026 and the last 500 days
- (a) Historical VaR and ES at $\alpha = 1\%$ and $\alpha = 2.5\%$
- (b) 95% bootstrap interval for $\text{VaR}_{1\%}$: 2000 resamples with replacement, percentile method [Efron \(1979\)](#)
- Code: `hs_var_es`, `boot_ci` in the notebook

Interpretation

- How many observations lie beyond $\text{VaR}_{1\%}$ in each sample?
- Which estimate would you report to a risk committee today, and why?
- What does the i.i.d. bootstrap ignore?

B1: Results [Solved]



	n	$\text{VaR}_{1\%}$	95% CI	$\text{ES}_{1\%}$	$\text{ES}_{2.5\%}$
2000–2026	6 670	4.13	[3.85, 4.46]	6.28	4.57
Last 500 days	500	2.71	[2.11, 3.12]	3.22	2.69

- Beyond $\text{VaR}_{1\%}$: 67 days in the full sample, only 5 in the last 500: the short-window ES is an average of 5 numbers
- The recent market is calmer: the two intervals do not overlap
- Report both: the recent value for today's risk, the long-run value as a stress reference
- The i.i.d. bootstrap ignores volatility clustering: the true intervals are wider (see B6)

B2: GPD Fit and Extreme-Value VaR on BET [Solved]

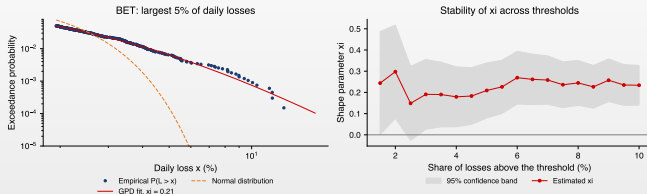
Task

- ▣ (a) Threshold u leaving the largest 5% of BET losses above it; fit a GPD to the excesses by maximum likelihood
- ▣ (b) Standard errors: $SE(\hat{\xi}) = (1 + \hat{\xi})/\sqrt{N_u}$,
 $SE(\hat{\beta}) = \hat{\beta}\sqrt{2(1 + \hat{\xi})/N_u}$
- ▣ (c) Extreme-value VaR and ES at $\alpha = 1\%$, 0.5% , 0.1% vs historical McNeil, Frey & Embrechts (2015)
- ▣ (d) Stability of $\hat{\xi}$ for thresholds leaving from 10% down to 1.5% of the losses above them

Interpretation

- ▣ Is the BET tail heavier than the Normal distribution? How sure are we?
- ▣ Where do the two methods agree, and where does only EVT give a usable number?
- ▣ How much does the answer depend on the threshold?

B2: Results [Solved]



- $u = 1.93\%$, $N_u = 334$ of $n = 6\,670$; $\hat{\xi} = 0.209$ (SE 0.066), $\hat{\beta} = 1.20$ (SE 0.10): $\hat{\xi}$ is 3 standard errors above 0, a heavy tail
- $\text{VaR}_{1\%} = 1.93 + \frac{1.20}{0.209} \left[\left(\frac{0.01}{0.050} \right)^{-0.209} - 1 \right] = 4.22\%$ (historical 4.13%)
- $\text{VaR } 0.5\%$: 5.47 vs 5.38; $\text{VaR } 0.1\%$: 9.18 vs 9.50 (only 7 observations beyond); $\text{ES}_{0.1\%}$: 12.61 vs 11.25
- $\hat{\xi}$ ranges from 0.15 to 0.30 across thresholds, but $\text{VaR}_{0.1\%}$ only from 9.04% to 9.51%: the quantile is more stable than the shape

B3: Filtered vs Historical Simulation on Bitcoin [Proposed]

Solution discussed in class

Task

- ▣ (a) Fit AR(1)-GARCH(1,1) to Bitcoin returns 2014–2026; forecast $\hat{\sigma}_{t+1}$ for 19 September 2026 [Barone-Adesi et al. \(1999\)](#)
- ▣ (b) Filtered $\text{VaR}_{1\%}$ and $\text{ES}_{1\%}$; compare with historical simulation on 500 days and on the full sample
- ▣ (c) Count the days in the last two years with a loss above each VaR (one-day-ahead forecasts)
- ▣ Model: B1

Interpretation

- ▣ Why can the three numbers in (b) differ by a factor of two?
- ▣ Which method misses more often, and are its misses clustered?
- ▣ What assumption of the filtered method would you test first?

B4: Cornish–Fisher vs Empirical Quantiles [Proposed]

Solution discussed in class

Task

- S&P 500, BET, Bitcoin, EUR/RON, gold: full samples
- (a) Sample skewness and excess kurtosis of the losses
- (b) VaR 1% and VaR 2.5%: Normal, Cornish–Fisher, historical, with a bootstrap interval for the historical VaR 1%
- Model: A1 (parametric), B1 (historical, bootstrap)

Interpretation

- Is the Cornish–Fisher value inside the bootstrap interval for any series?
- Why does it work better for VaR 2.5% than for VaR 1%?
- Would it help to use moments of the last year only?

B5: Component VaR of a Four-Asset Portfolio [Proposed]

Solution discussed in class

Task

- Equal weights (25% each) in SPY, TLT, GLD and Bitcoin; daily simple returns, September 2014 – September 2026
- (a) Normal $\text{VaR}_{1\%}$, marginal and component VaR (Euler) Tasche (2008)
- (b) Historical $\text{ES}_{2.5\%}$ contributions $\mathbb{E}[-w_i R_i \mid L_p \geq \text{VaR}_{2.5\%}]$
- Model: A5

Interpretation

- Is an equally weighted portfolio equally risky in each asset?
- Compare with the 40/30/20/10 portfolio of the lecture
- Which weights would equalise the risk contributions (risk parity)?

B6: How Precise Is an ES Estimate? [Proposed]

Solution discussed in class

Task

- ▣ Historical $ES_{2.5\%}$ for the S&P 500 and BET: full sample and last 500 days
- ▣ (a) 95% percentile interval by the i.i.d. bootstrap [Efron \(1979\)](#), 2 000 resamples
- ▣ (b) Same with the moving-block bootstrap, blocks of 20 days [Künsch \(1989\)](#)
- ▣ Model: B1

Interpretation

- ▣ How many observations does each ES average?
- ▣ Why is the block interval wider on the full sample?
- ▣ Can you tell apart two ES models that differ by 0.3 percentage points with 500 days [Pele & Mazurencu-Marinescu-Pele \(2026\)](#)?

C1: Capital for a BVB Portfolio vs an S&P 500 Portfolio

Question

- How much capital does ES 2.5% require for 1 million invested in a BVB blue-chip portfolio versus an S&P 500 portfolio?
- BVB: Banca Transilvania, OMV Petrom, BRD, Transgaz, Romgaz, Nuclearelectrica, Electrica, Transelectrica, equal weights; S&P 500: SPY; last ten years
- Follow the FRTB logic: 10-day horizon, a stressed period, liquidity horizons [Basel Committee \(2019\)](#)

What a good answer contains

- One-day and 10-day ES: $\sqrt{10}$ scaling vs overlapping 10-day losses vs filtered Monte Carlo
- The most severe 12-month window and the ES inside it
- Are BVB blue chips “large cap” (10 days) or “small cap” (20 days)?
- Thin trading, currency (RON vs USD) and the uncertainty of every number

C2: Critique an AI Answer [Proposed]

Solution discussed in class

The prompt sent to an artificial intelligence (AI) assistant

- “px holds daily BET closes, 2000–2026. Give me the one-day historical VaR 1% and ES 2.5% as positive losses in %, and the 10-day VaR 1%.”

The assistant's answer

```
import numpy as np
r = 100 * np.log(px).diff().dropna()           # daily log returns, %
loss = -r                                       # loss in % of the position
var1 = np.percentile(r, 99)                   # VaR 1%
es25 = loss[loss >= np.percentile(loss, 99)].mean() # ES 2.5%
var10 = 10 * var1                             # 10-day VaR 1%
```

- “VaR 1% = 3.95%, ES 2.5% = 6.28%, 10-day VaR 1% = 39.46% of the position.”

C2: Critique an AI Answer, the Task [Proposed]

Solution discussed in class

- The code runs, but the answer contains **three** mistakes typical for VaR and ES
 - ▶ Everything else in the answer is correct
- For each mistake
 - ▶ Name the line and say why it is wrong
 - ▶ Say how you would detect it: a count of observations, a number from B1, a slide of the lecture
 - ▶ Write the corrected line(s) and the corrected numbers
- Log each mistake as in `AI_ERRORS.md`
 - ▶ What was asked, what the AI answered, what was wrong, how you found out, the correct version
- Hint: how many daily losses does `es25` average, and how many should it average?

Common Pitfalls

- ▣ Mixing sign conventions: a VaR reported as a negative return in one table and a positive loss in another
- ▣ Using the Normal distribution or Cornish–Fisher with daily kurtosis of 10
- ▣ Reporting ES 1% from 500 days without saying it averages 5 numbers
- ▣ Scaling by $\sqrt{10}$ a series with positive autocorrelation (BET, the BVB basket)
- ▣ Reading money weights as risk weights: a 10% crypto position can carry over a third of the risk
- ▣ Log returns summed across assets: portfolio losses need simple returns

Take-Home Assignment (Team Project)

- Choose a portfolio of at least three assets not covered in class (BVB stocks, a currency, a crypto-asset)
- Estimate one-day $\text{VaR}_{1\%}$ and $\text{ES}_{2.5\%}$ with historical, filtered and extreme-value methods, each with a bootstrap interval
- Split the ES into Euler contributions and propose weights that reduce ES by 20% at the same expected return
- Log every AI error you catch in `AI_ERRORS.md` (at least three per project)
- Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Measuring Statistical Risk
- ▣  XFG Advanced Methods in Quantitative Finance

Code and data

- ▣ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_07

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